

Title : PCB DEFECT DETECTION USING CONDITIONAL DCGAN

TEAM 10

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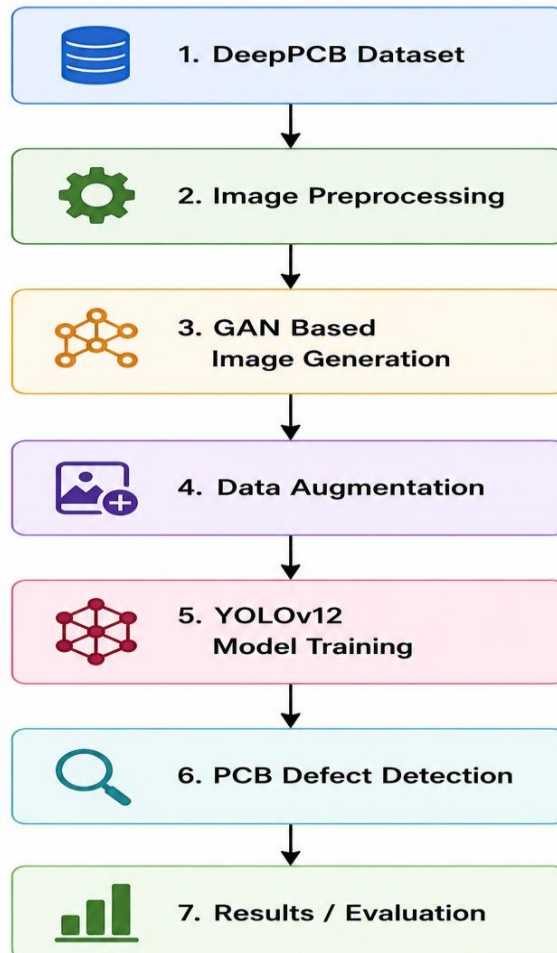
Problem Statement

Detecting PCB defects manually is difficult, time-consuming, and requires large amounts of defect data for training AI models. Existing datasets are limited and imbalanced. This project uses GAN to generate synthetic PCB defect images and YOLO for accurate PCB defect detection.

Proposed System

- Deep PCB dataset is used
- Image preprocessing is performed
- GAN is used to generate synthetic PCB defect images
- Data augmentation is applied
- YOLOv12 model is trained using real and synthetic images
- PCB defects are detected automatically
- Performance is evaluated using Precision, Recall, and mAP metrics

Detailed Design of Proposed System



- Deep PCB dataset contains different PCB defect images.
- Image preprocessing is done before training.
- Conditional DCGAN is used to generate synthetic PCB defect images.
- Data augmentation improves dataset balance and diversity.
- YOLOv12-S model is trained using real and generated images.
- PCB defects are detected automatically using the trained model.
- Precision, Recall and mAP metrics are used for performance evaluation.

Components and Their Relationships

- Deep PCB dataset contains different PCB defect images.
- Image preprocessing is done before training.
- Conditional DCGAN is used to generate synthetic PCB defect images.
- Data augmentation improves dataset balance and diversity.
- YOLOv12-S model is trained using real and generated images.
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Programming and EDA Tools Used

Programming Languages

- Python

Deep Learning Models

- Conditional DCGAN
- YOLOv12-S

Libraries Used

- OpenCV
- NumPy
- Matplotlib
- PyTorch

Development Platform

- Google Colab

Dataset

- Deep PCB Dataset

Evaluation Metrics

- Precision : 98.63%
- Recall : 95.28%
- mAP@50 : 98.00%
- mAP@50-95 : 69.48%

Implementation

- Deep PCB dataset is collected.
- Image preprocessing is performed.
- Labels are extracted from TXT files.
- Conditional DCGAN generates synthetic PCB defect images.
- Data augmentation is applied to improve the dataset.
- YOLOv12-S model is trained using real and synthetic images.
- PCB defects are detected using the trained model.
- Performance is evaluated using Precision, Recall, and mAP metrics.

Challenges

- Limited PCB defect dataset
- GAN training instability
- Class imbalance in dataset
- High training time

Solutions

- Synthetic images generated using Conditional DCGAN
- Data augmentation is applied
- Parameter tuning is performed
- YOLOv12-S model is trained using balanced data

Task Completion / Project Progress

- Dataset collection completed
- Image preprocessing completed
- Synthetic image generation completed
- YOLOv12-S model training completed
- PCB defect detection completed
- Performance evaluation completed

THANK YOU